

Pseudocontexts forced by finite context hypergraphs

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We give a restrictive definition of a pseudocontext. Two disjoint families of propositions form a pseudocontext only when their equality of probability sums is forced by finitely many context-normalization equations. The definition is therefore independent of a chosen coordinatization and of the mechanism used to generate probabilities, provided mutually exclusive propositions are additive and every context is complete. For the incidence matrix M of a finite context hypergraph and $d = \chi_A - \chi_B$, forcing is equivalent to the existence of a balanced certificate λ satisfying $M^T \lambda = d$ and $\mathbf{1}^T \lambda = 0$. This supplies an exact rational algorithm and includes complementary context coverings as integral certificates. We apply the criterion to a 15-vertex real three-dimensional construction with pseudocontexts of size three and to a 20-vertex complex three-dimensional construction with pseudocontexts of size two. The latter is the minimum possible size over $\mathbb{C}$, whereas size three is minimal over $\mathbb{R}$. Equal projector sums that hold only for selected frames or coordinates are distinguished as representation-dependent identities and are not pseudocontexts in the present sense.

I. INTRODUCTION

A context is a complete collection of mutually exclusive propositions. If p is a probability assignment and E is a context, completeness and finite additivity give

$$\sum_{v \in E} p(v) = 1. \quad (1)$$

This elementary equation is shared by classical sample-space models, quantum probabilities on orthogonal bases, and any other normalized model of the same propositions. It is therefore natural to ask whether finitely many equations of the form (1) can force an additive identity between two families that are not themselves contexts.

Such families were called *pseudocontexts* in Ref. [1]. There is, however, a necessary distinction. For particular rays a_i, b_i in a Hilbert space, one may happen to have

$$\sum_i |a_i\rangle\langle a_i| = \sum_i |b_i\rangle\langle b_i|. \quad (2)$$

Equation (2) implies equality of Born-probability sums for those coordinates, but the equality may disappear when the same orthogonality hypergraph is coordinatized differently. It also says nothing about probability models that are not derived from those projectors.

Here we reserve *pseudocontext* for the stronger, finite-combinatorial notion: the equality must be a balanced linear consequence of the context equations themselves. This implements two requirements simultaneously:

1. the identity holds for every faithful orthogonal representation of the hypergraph, not merely for one selected vector realization;
2. the identity holds for every admissible probability assignment, independently of whether it is classical, quantum, softmax-generated [2], or obtained by another rule.

Only completeness, additivity on mutually exclusive propositions, and a single context-independent weight for each vertex are used.

The central result is an incidence-matrix characterization. Besides making the definition testable, it cleanly separates three issues that can otherwise be conflated: combinatorial forcing, geometric realizability, and the state space of a particular probability theory. We then revisit the three-dimensional constructions of Ref. [1] and a 20-vertex octagon whose scalar-field representability is analyzed separately in Ref. [3].

II. CONTEXTS, WEIGHTS, AND ORTHOGONAL REPRESENTATIONS

Let $\Gamma = (V, \mathcal{E})$ be a finite n -uniform hypergraph. Its vertices are elementary propositions and every $E \in \mathcal{E}$ is declared to be a complete context of n mutually exclusive propositions. We exclude repeated vertices inside a context.

Definition 1 (Admissible weight). An *admissible weight* on Γ is a function $p : V \rightarrow [0, 1]$ satisfying Eq. (1) for every $E \in \mathcal{E}$. The set of admissible weights is denoted by $\mathcal{S}(\Gamma)$. The hypergraph is *probabilistically consistent* if $\mathcal{S}(\Gamma) \neq \emptyset$.

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The range $[0,1]$ records the probabilistic interpretation, but the forcing criterion below is deliberately stronger: it requires an identity on the entire real affine solution space of the normalization equations. Positivity will therefore play no part in the proof.

Different theories select different subsets of $\mathcal{S}(\Gamma)$. Classical deterministic models use two-valued weights and their convex combinations. A quantum realization assigns rays to the vertices and uses the Born rule. A softmax, neural, or statistical parametrization is also covered whenever its output is a single, globally defined probability assignment $p : V \rightarrow [0,1]$ satisfying every context-normalization equation; that is,

$$\sum_{v \in E} p(v) = 1 \quad \text{for every } E \in \mathcal{E}.$$

Any probability assignment that is not a single globally defined function on V —so that a vertex shared by different contexts may receive different values in those contexts—is not a weight on Γ . It instead describes context-labelled outcomes with context-dependent frequencies or expectation values.

To distinguish properties of the abstract hypergraph from properties of a particular vector or projector realization, let $\mathbb{P}(H)$ denote the set of rays, that is, the one-dimensional subspaces, of an n -dimensional real or complex Hilbert space H . A context $E \in \mathcal{E}$ is an n -element subset of vertices; it is not itself a projection. We use the faithful version of the standard graph-theoretic notion of an orthogonal representation [4–6].

Definition 2 (Faithful orthogonal representation). A *faithful orthogonal representation* (FOR), also called a *faithful coordinatization*, of $\Gamma = (V, \mathcal{E})$ in H is an injective map

$$h : V \rightarrow \mathbb{P}(H).$$

For each vertex $v \in V$, choose a unit vector $x_v \in H$ spanning the ray $h(v)$. The representation is called faithful if, for all distinct $u, v \in V$,

$$\begin{aligned} \langle x_u, x_v \rangle = 0 & \iff h(u) \perp h(v) \\ & \iff \text{there is an } E \in \mathcal{E} \\ & \quad \text{such that } \{u, v\} \subseteq E. \end{aligned} \quad (3)$$

The condition is independent of the chosen unit representatives, since changing a representative by a sign or a complex phase does not affect whether the inner product is zero.

Since $|E| = n = \dim H$, the vectors $\{x_v : v \in E\}$ form an orthonormal basis of H for every $E \in \mathcal{E}$. Equivalently, define

$$P_{h(v)} = |x_v\rangle\langle x_v|.$$

This is the rank-one orthogonal projection onto the ray $h(v)$; in particular, it is self-adjoint and idempotent. For

every context $E \in \mathcal{E}$, the corresponding projections are pairwise orthogonal and satisfy

$$\sum_{v \in E} P_{h(v)} = I_H. \quad (4)$$

The hypergraph is *geometrically admissible* in H if such a faithful orthogonal representation exists.

III. RESTRICTIVE DEFINITION AND CERTIFICATE THEOREM

Index the rows by $\mathcal{E}$ and the columns by V , and let

$$M = M_\Gamma \in \{0,1\}^{\mathcal{E} \times V}, \quad M_{Ev} = \begin{cases} 1, & v \in E, \\ 0, & v \notin E, \end{cases} \quad (5)$$

be the incidence matrix of Γ . Thus, for $p \in \mathbb{R}^V$ and $E \in \mathcal{E}$,

$$(Mp)_E = \sum_{v \in V} M_{Ev} p(v) = \sum_{v \in E} p(v).$$

Consequently, the system

$$Mp = \mathbf{1}_\mathcal{E} \quad (6)$$

means exactly that

$$\sum_{v \in E} p(v) = 1 \quad \text{for every } E \in \mathcal{E}.$$

For $A \subseteq V$, let $\chi_A \in \{0,1\}^V$ denote its incidence vector.

Definition 3 (Balanced context certificate). Let $A, B \subseteq V$ be disjoint. A vector $\lambda \in \mathbb{Q}^\mathcal{E}$ is a *balanced context certificate* for (A, B) if

$$M^\top \lambda = \chi_A - \chi_B, \quad \mathbf{1}_\mathcal{E}^\top \lambda = 0. \quad (7)$$

The first equation means that, in the linear combination of context equations with coefficients λ_E , every vertex outside $A \cup B$ cancels, while vertices in A and B occur with coefficients $+1$ and -1 , respectively. The second equation ensures that the constant right-hand sides of the context equations cancel as well. Therefore, for every solution of $Mp = \mathbf{1}_\mathcal{E}$,

$$\sum_{a \in A} p(a) - \sum_{b \in B} p(b) = (\chi_A - \chi_B)^\top p = \lambda^\top Mp = \lambda^\top \mathbf{1}_\mathcal{E} = 0.$$

Hence a balanced context certificate forces

$$\sum_{a \in A} p(a) = \sum_{b \in B} p(b).$$

a. Example (balanced context: firefly logic). The following elementary five-vertex configuration is the standard firefly logic [7]. Let

$$V = \{a, b, c, d, x\}, \quad E_1 = \{a, b, x\}, \quad E_2 = \{c, d, x\}.$$

A real three-dimensional coordinatization is given by $a = (1, 0, 0)$, $b = (0, 1, 0)$, $x = (0, 0, 1)$, and $c = (1/\sqrt{2})(1, 1, 0)$, $d = (1/\sqrt{2})(1, -1, 0)$. Thus E_1 and E_2 are both orthonormal bases, and the corresponding normalization equations are

$$p(a) + p(b) + p(x) = 1, \quad p(c) + p(d) + p(x) = 1.$$

Subtracting the second equation from the first gives

$$p(a) + p(b) = p(c) + p(d).$$

This equality is expressed by the balanced context certificate

$$A = \{a, b\}, \quad B = \{c, d\}, \quad \lambda = (1, -1)^\top.$$

Indeed, ordering the vertices as (a, b, c, d, x) , the incidence matrix is

$$M = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{pmatrix},$$

and hence

$$M^\top \lambda = (1, 1, -1, -1, 0)^\top = \chi_A - \chi_B, \\ \mathbf{1}_\mathcal{E}^\top \lambda = 1 - 1 = 0.$$

The common vertex x cancels from the two sides, while the constants on the right-hand sides cancel because the two context equations are taken with opposite coefficients. This is a balanced context in the present terminology, but not a pseudocontext under Definition 4: the two elements $a, b \in A$ occur together in the context E_1 , and the two elements $c, d \in B$ occur together in the context E_2 . Thus Condition 2 is violated on both sides.

Definition 4 (Pseudocontext). Let $k \geq 2$. A *pseudocontext of size k* is a triple $(\Gamma; A, B)$ such that

1. Γ is a finite, probabilistically consistent, uniform context hypergraph;
2. $A, B \subseteq V$ are disjoint k -element sets such that no context contains two distinct elements of A and no context contains two distinct elements of B ;
3. (A, B) has a balanced context certificate.

The restriction $k \geq 2$ excludes the elementary size-one identities obtained by subtracting context-normalization equations. Such identities remain balanced context certificates, but are not called pseudocontexts here. It is a *pseudocontext in H* if, in addition, Γ admits a FOR in H .

Condition 2 removes ordinary contexts and their incomplete mutually exclusive subsets. Under a FOR, each target family is pairwise nonorthogonal. Disjointness prevents identities obtained by padding both sides with the same propositions.

Theorem 5 (Finite forcing criterion). *Suppose the real affine system $Mp = \mathbf{1}_\mathcal{E}$ is nonempty and let $d = \chi_A - \chi_B$. The following are equivalent:*

1. $d^\top p = 0$ is a linear consequence of the context equations, that is, it holds for every real solution of $Mp = \mathbf{1}_\mathcal{E}$;
2. there is a real vector λ such that $M^\top \lambda = d$ and $\mathbf{1}_\mathcal{E}^\top \lambda = 0$;
3. there is a rational balanced context certificate for (A, B) .

Whenever these conditions hold,

$$\sum_{a \in A} p(a) = \sum_{b \in B} p(b) \quad (8)$$

for every admissible weight on Γ .

Proof. If 2 holds and $Mp = \mathbf{1}_\mathcal{E}$, then

$$d^\top p = \lambda^\top Mp = \lambda^\top \mathbf{1}_\mathcal{E} = 0,$$

so 2 implies 1 and Eq. (8) follows.

Conversely, choose one solution p_0 of $Mp = \mathbf{1}_\mathcal{E}$. Every vector $x \in \ker M$ gives another solution $p_0 + x$. By 1, $d^\top(p_0 + x) = d^\top p_0 = 0$, hence d annihilates $\ker M$. Finite-dimensional linear algebra gives $d \in (\ker M)^\perp = \text{im } M^\top$, so $d = M^\top \lambda$ for some real λ . Evaluating at p_0 yields $0 = d^\top p_0 = \lambda^\top \mathbf{1}_\mathcal{E}$, proving 2.

The equations in 2 have integer coefficients. A consistent linear system over $\mathbb{Q}$ has a rational solution, for example by Gaussian elimination. Thus 2 and 3 are equivalent. $\square$

The word “forced” in Definition 4 is therefore literal: the identity is valid even before nonnegativity or a particular state model is imposed. If one only asks for equality on the polytope $\mathcal{S}(\Gamma)$, additional equalities may arise because positivity forces certain vertices to zero. Those boundary effects are intentionally excluded from the definition.

Corollary 6 (Integral form). *Every pseudocontext admits an integer $q \geq 1$ and $\mu \in \mathbb{Z}^\mathcal{E}$ such that*

$$M^\top \mu = q(\chi_A - \chi_B), \quad \mathbf{1}_\mathcal{E}^\top \mu = 0. \quad (9)$$

Proof. Clear the denominators of a rational certificate. Applying an admissible real-valued weight to Eq. (9) gives q times the desired equality, and division by q completes the argument. $\square$

Corollary 7 (Independence of coordinatization). *Let $(\Gamma; A, B)$ be a pseudocontext and let h be any FOR of Γ in an n -dimensional real or complex Hilbert space. Then*

$$\sum_{a \in A} P_{h(a)} = \sum_{b \in B} P_{h(b)}. \quad (10)$$

Consequently the Born-probability sums are equal for every density operator, independently of the chosen FOR.

Proof. Multiply each operator equation (4) by the corresponding certificate coefficient and sum over $E \in \mathcal{E}$. The left-hand side becomes $\sum_v (M^\top \lambda)_v P_{h(v)}$, while the right-hand side is $(\mathbf{1}^\top \lambda) I_H = 0$. Equation (7) gives Eq. (10). $\square$

This proof does not invoke Gleason's theorem. Gleason's theorem [8] characterizes quantum states in dimensions at least three, but the pseudocontext equality already follows from the finite context equations.

IV. COVERING CERTIFICATES AND EXACT DISCOVERY

The most transparent certificates compare two multi-sets of contexts. Suppose $\mathcal{F}_+$ and $\mathcal{F}_-$ contain the same number of contexts, counted with multiplicity. If their vertex multiplicities agree away from $A \cup B$ and their residual incidence difference is $\chi_A - \chi_B$, then assigning $+1$ to $\mathcal{F}_+$, -1 to $\mathcal{F}_-$, and zero elsewhere gives Eq. (7). We call this a *complementary covering certificate*.

The general criterion is no harder to test. For fixed A, B , exact rational row reduction of

$$\begin{pmatrix} M^\top \\ \mathbf{1}_\mathcal{E}^\top \end{pmatrix} \lambda = \begin{pmatrix} \chi_A - \chi_B \\ 0 \end{pmatrix} \quad (11)$$

either returns a certificate or proves that none exists. To discover pseudocontexts rather than verify a proposed pair, compute the consequence space

$$\mathcal{C}_\Gamma = M^\top \{\lambda \in \mathbb{Q}^\mathcal{E} : \mathbf{1}^\top \lambda = 0\} \quad (12)$$

and search it for sparse vectors with entries in $\{-1, 0, 1\}$. The positive and negative supports of such a vector are candidate target families; the nonexclusivity and disjointness conditions are then checked combinatorially. Integer programming can be used to minimize the number of contexts or vertices in a certificate.

For an n -uniform hypergraph, every $d \in \mathcal{C}_\Gamma$ satisfies $\mathbf{1}_V^\top d = 0$, because

$$\mathbf{1}_V^\top M^\top \lambda = n \mathbf{1}_\mathcal{E}^\top \lambda = 0.$$

Thus a $\{-1, 0, 1\}$ consequence automatically has equally large positive and negative supports. Equal pseudocontext size is not an extra numerical coincidence; it follows from uniformity and balance.

V. A SIZE-THREE REAL PSEUDOCONTEXT

Consider the 15-vertex, eight-context hypergraph introduced in this role in Ref. [1]. Its contexts are

$$\begin{aligned} E_1 &= \{1, 2, 3\}, & E_2 &= \{3, 4, 5\}, \\ E_3 &= \{6, 7, 8\}, & E_4 &= \{8, 9, 10\}, \\ E_5 &= \{11, 12, 13\}, & E_6 &= \{13, 14, 15\}, \\ E_7 &= \{2, 7, 12\}, & E_8 &= \{4, 9, 14\}. \end{aligned} \quad (13)$$

Let

$$A = \{1, 6, 11\}, \quad B = \{5, 10, 15\}. \quad (14)$$

No two vertices within either triple share a context.

The four contexts

$$\mathcal{F}_+ = \{E_1, E_3, E_5, E_8\} \quad (15)$$

cover every vertex once except B , whereas

$$\mathcal{F}_- = \{E_2, E_4, E_6, E_7\} \quad (16)$$

cover every vertex once except A . Therefore

$$\sum_{j \in \{1, 3, 5, 8\}} \chi_{E_j} - \sum_{j \in \{2, 4, 6, 7\}} \chi_{E_j} = \chi_A - \chi_B. \quad (17)$$

The two sides use four contexts each, so the certificate is balanced. Every admissible weight consequently satisfies

$$p(1) + p(6) + p(11) = p(5) + p(10) + p(15). \quad (18)$$

This hypergraph has classical partition representations and real three-dimensional FORs [1]. Hence Eq. (18) holds in its classical models, in every one of its quantum FORs, and for any other admissible weight. The numerical range of the common sum may still depend on the chosen theory; the equality itself does not.

VI. A SIZE-TWO COMPLEX PSEUDOCONTEXT

A smaller target family is forced by a 20-vertex, 12-context hypergraph. Let $v_0, \dots, v_7$ be the midpoints of a cyclic octagon and $v_{01}, v_{12}, \dots, v_{70}$ its corners, with indices modulo eight. The eight perimeter contexts are

$$E_i = \{v_{i-1, i}, v_i, v_{i, i+1}\}, \quad i \in \mathbb{Z}_8. \quad (19)$$

Four further contexts are

$$\begin{aligned} D_0 &= \{v_0, a_1, v_4\}, & D_2 &= \{v_2, a_2, v_6\}, \\ D_1 &= \{v_1, b_2, v_5\}, & D_3 &= \{v_3, b_1, v_7\}. \end{aligned} \quad (20)$$

Take $A = \{a_1, a_2\}$ and $B = \{b_1, b_2\}$. The six contexts

$$\begin{aligned} \mathcal{G}_+ &= \{D_0, D_2, E_1, E_3, E_5, E_7\}, \\ \mathcal{G}_- &= \{D_1, D_3, E_0, E_2, E_4, E_6\} \end{aligned} \quad (21)$$

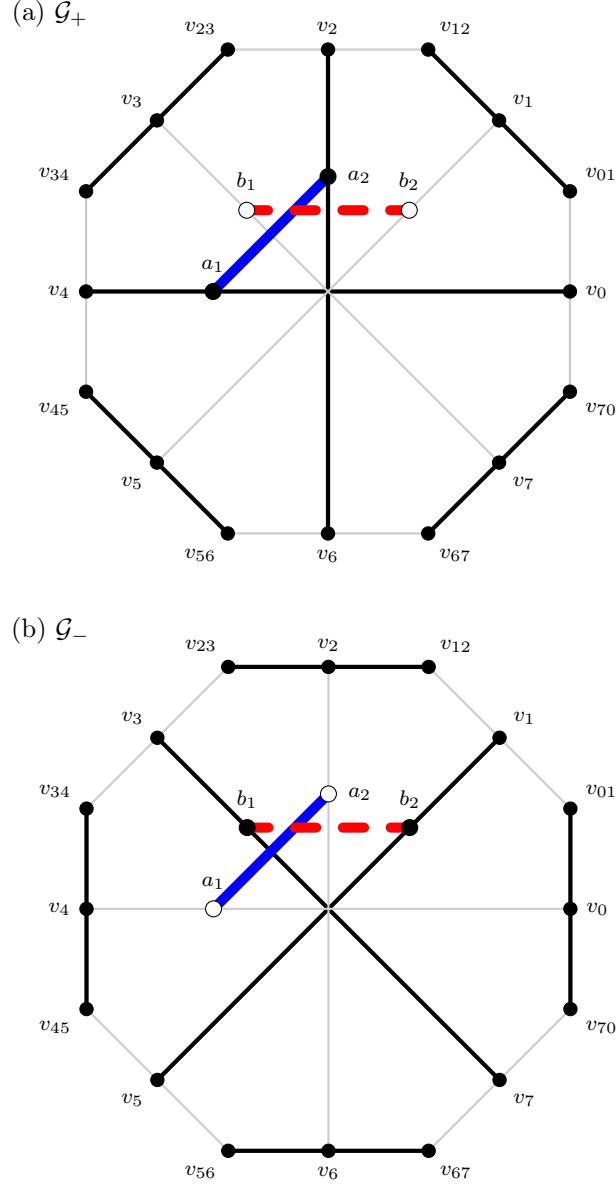


FIG. 1. The two context coverings used in the incidence-vector certificate. Bold black lines are the selected contexts and thin gray lines are the unselected contexts. Panel (a) shows $\mathcal{G}_+$ and panel (b) shows $\mathcal{G}_-$. In both panels, the very thick blue solid line joins $A = \{a_1, a_2\}$ and the very thick red dashed line joins $B = \{b_1, b_2\}$. Filled inner vertices are covered by the respective covering.

each cover all 16 outer vertices once. The first also covers a_1, a_2 , while the second covers b_1, b_2 . The two coverings are shown in Fig. 1.

Subtraction gives the balanced certificate

$$\sum_{G \in \mathcal{G}_+} \chi_G - \sum_{G \in \mathcal{G}_-} \chi_G = \chi_{a_1} + \chi_{a_2} - \chi_{b_1} - \chi_{b_2}. \quad (22)$$

Thus every admissible weight satisfies

$$p(a_1) + p(a_2) = p(b_1) + p(b_2). \quad (23)$$

The same equality also has a direct two-cover proof.

Let

$$S_{\text{out}}(p) = \sum_{i=0}^7 p(v_i) + \sum_{i=0}^7 p(v_{i+1}),$$

where the corner indices are read modulo eight. Adding the six normalization equations belonging to the first cover gives

$$6 = S_{\text{out}}(p) + p(a_1) + p(a_2), \quad (24)$$

whereas adding those belonging to the second cover gives

$$6 = S_{\text{out}}(p) + p(b_1) + p(b_2). \quad (25)$$

The outer-vertex contribution is identical because each cover contains every outer vertex exactly once. Subtracting these two equations recovers Eq. (23). Thus the same pseudocontext identity is visible both as the incidence-vector certificate and as the difference of two complete covers.

The forcing proof contains no coordinates and no scalar field. Realizability is nevertheless field-sensitive. The hypergraph has an exact FOR in $\mathbb{C}^3$, while it has no FOR in $\mathbb{R}^3$; exact coordinates and the obstruction are proved in Ref. [3]. It is therefore a pseudocontext in $\mathbb{C}^3$, but not in $\mathbb{R}^3$. This example illustrates why forcing and geometric admissibility must be stated as separate conditions.

VII. MINIMUM SIZES OVER THE REAL AND COMPLEX FIELDS

The preceding examples give sharp minimum sizes for faithful Hilbert-space pseudocontexts under the present definition.

Proposition 8 (No size-one Hilbert-space pseudocontext). *There is no pseudocontext of size one in a real or complex Hilbert space.*

Proof. If $A = \{a\}$ and $B = \{b\}$ had a balanced certificate, then Corollary 7 would give $P_{h(a)} = P_{h(b)}$ in every FOR. Equality of rank-one projectors means equality of rays, contradicting injectivity of h because $a \neq b$. $\square$

Proposition 9 (Real rigidity at size two). *There is no pseudocontext of size two in a real Hilbert space whose two target families are internally nonorthogonal.*

Proof. Suppose unit vectors a_1, a_2, b_1, b_2 in a real Hilbert space satisfy

$$P_{a_1} + P_{a_2} = P_{b_1} + P_{b_2} =: T.$$

The range of T is the common two-dimensional span of both pairs. Change the sign of a_2 if necessary so that $c = \langle a_1, a_2 \rangle > 0$. In an eigenbasis of T its eigenvalues are $1 + c$ and $1 - c$, which are distinct. Write the two unit vectors of any decomposition in this eigenbasis as $(\cos \phi, \sin \phi)$ and $(\cos \psi, \sin \psi)$. The diagonal and off-diagonal equations combine to $e^{2i\phi} + e^{2i\psi} = 2c$. Two points on the unit circle with this sum are the conjugate pair $c \pm i\sqrt{1 - c^2}$. The two rays are therefore uniquely determined, up to interchange, by T . Hence $\{[b_1], [b_2]\} = \{[a_1], [a_2]\}$, contradicting disjointness.

If $c = 0$, then the first pair is orthogonal, contrary to the internal nonorthogonality condition. Thus no allowed real size-two pair exists. $\square$

Complex phases remove the real uniqueness used in the proof. The octagon of Section VI shows that this phase freedom can be enforced by a finite context hypergraph rather than merely arranged in selected coordinates.

Theorem 10 (Sharp field-dependent threshold). *The minimum size of a pseudocontext admitting a FOR over $\mathbb{C}$ is 2. The minimum size over $\mathbb{R}$ is 3.*

Proof. Proposition 8 gives the universal lower bound. The octagon with a FOR from Section VI attains size two over $\mathbb{C}^3$. Proposition 9 excludes size two over $\mathbb{R}$, while the construction of Section V attains size three in $\mathbb{R}^3$. $\square$

VIII. WHAT THE DEFINITION EXCLUDES

The stronger definition changes the classification of several familiar operator identities. If two unit-norm frames decompose the same positive operator, Eq. (2) holds and all Born states give the same sum. Tight frames provide many such examples: a frame $(a_i)_{i=1}^k$ with

$$\sum_i P_{a_i} = \frac{k}{n} I$$

and a generic rotation of it yield distinct decompositions of the same operator. Likewise, in $\mathbb{C}^2$ relative phases give distinct two-ray decompositions of one nonscalar positive operator.

These identities are *representation-dependent*. They become pseudocontexts only if a finite auxiliary context hypergraph supplies a certificate of the form (7). The bare rays of a generic rotated tight-frame pair may have no contexts at all, in which case its incidence matrix cannot force a nonzero target difference. An operator identity is thus a necessary consequence of a pseudocontext realized by a FOR, by Corollary 7, but it is not a sufficient definition.

This distinction also clarifies the role of the algebraic formalization of rank-one sums. Coordinate calculations, including machine-checked calculations, can verify Eq. (2), trace identities, ray distinctness, and nonorthogonality for a proposed realization. The additional pseudocontext claim is certified separately and exactly by the finite incidence equation (7).

Ordinary contexts are normalized independently of both coordinatization and probability model. Pseudocontexts inherit that same twofold independence through a balanced finite context certificate. A bare projector-sum identity, by contrast, concerns Born-rule states for the chosen rank-one decomposition and is generally coordinatization-dependent.

IX. OPERATIONAL AND CONTEXTUALITY REMARKS

Members of one pseudocontext need not be jointly measurable. The expression $\sum_{a \in A} p(a)$ is a linear statistic obtained from separate, mutually incompatible measurement arrangements and repeated preparations, not

the probability of one coarse-grained event. It can exceed one. The certificate asserts that the same statistic is obtained from B for every admissible preparation.

The equality is not by itself a contextuality witness. Indeed it holds in every classical two-valued state whenever such states exist, as well as in their convex mixtures. Classical and quantum models may nevertheless allow different ranges for the common sum, as in the examples of Ref. [1]. Any contextuality claim must be based on that additional separation of state spaces, not on Eq. (8) alone. This is also distinct from generalized operational contextuality in the sense of Spekkens [9], where equivalences among arbitrary preparations, transformations, or measurement procedures enter the noncontextuality assumption.

The present equality is best regarded as a theory-independent conservation law generated by a finite network of complete tests. A probability model may refine the network, restrict its state space, or supply a geometric realization, but it cannot change a balanced incidence consequence while retaining the same vertex identifications and context equations.

X. DISCUSSION AND OPEN PROBLEMS

Definition 4 makes pseudocontexts finite, exact, and portable across probability theories. It also suggests several concrete problems.

First, one may classify the sparse signed vectors in $\mathcal{C}_\Gamma$ and minimize their context certificates. This is an integer-linear problem at the combinatorial level. Second, among the resulting abstract pseudocontexts, one may ask which admit FORs in $\mathbb{R}^n$, in $\mathbb{C}^n$, or in both. The octagon shows that these answers can differ even though the certificate is field-free. Third, one may compare the ranges of the common statistic over classical, quantum, and more general admissible state spaces. Only this third layer can produce a statistical separation once the common equality has been fixed.

A further structural question is whether every rational certificate can be replaced, perhaps after a controlled finite extension of the hypergraph, by a $\{-1, 0, 1\}$ complementary covering certificate. Such a result would turn ev-

ery abstract linear proof into a directly visible pair of coverings. Minimal extensions that convert a representation-dependent projector identity into a pseudocontext are another natural target.

XI. CONCLUSION

A pseudocontext should not depend on lucky coordinates. Under the restrictive definition adopted here, the equality of its two probability sums must be forced by finitely many completeness and additivity equations. The balanced incidence certificate $M^\top \lambda = \chi_A - \chi_B$, $\mathbf{1}^\top \lambda = 0$ is necessary and sufficient for precisely this notion of forcing. It proves the equality simultaneously for classical, quantum, softmax-generated, and other admissible weights, and it transfers to an operator identity in every Hilbert-space FOR.

The 15-vertex and 20-vertex examples show that this stronger concept is nonempty and retains a sharp real-complex minimum-size distinction. The separate questions “Is the equality forced?”, “Does the hypergraph admit a FOR?”, and “Which probability ranges does a theory allow?” should remain separate in subsequent applications.

DATA AVAILABILITY

All context lists and certificates needed for the results are displayed in the text. Exact coordinates and faithfulness certificates for the octagon are given in Ref. [3]. Lean 4 sources checking the finite coordinate-operator identities discussed in Section VIII are available from the corresponding authors upon reasonable request.

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